

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Easy

Question

	1 visit	2 or more visits	Total
Less than 40 years old	15	15	30
At least 40 years old	20	85	105
Total	35	100	135

The table summarizes customers who visited a car dealership in the last month by age and number of visits they made to the dealership. If a customer from the last month is selected at random, what is the probability that the selected customer is at least **40** years old?

Answer

- A. $\frac{30}{135}$
- B. $\frac{35}{135}$
- C. $\frac{100}{135}$
- D. $\frac{105}{135}$

Correct Answer: D

Rationale

Choice D is correct. Based on the table, there are a total of **135** customers who visited the car dealership in the last month, and **105** of these customers are at least **40** years old. If a customer from the last month is selected at random, the probability that the selected customer is at least **40** years old is equal to the number of customers who are at least **40** years old divided by the total number of customers. Therefore, the probability that the selected customer is at least **40** years old is $\frac{105}{135}$.

Choice A is incorrect. This is the probability that the selected customer is less than **40** years old.

Choice B is incorrect. This is the probability that the selected customer visited the dealership **1** time in the last month.

Choice C is incorrect. This is the probability that the selected customer visited the dealership **2** or more times in the last month.